

Gaussian Process Modeling and Missing-Data Imputation in Kepler Stellar Light Curves

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Abstract

Stellar light curves frequently suffer from missing observations and varying levels of stochastic and structured noise. This paper investigates the efficacy of Gaussian Process (GP) regression for modeling noisy stellar variability, imputing missing data, and recovering structured periodic behavior in Kepler mission data. Through a rigorous Monte Carlo evaluation utilizing 20 repeated randomized train-test splits, we benchmarked GP regression against linear interpolation and a classical time-series model. Our findings demonstrate that both Gaussian Process regression and ARIMA substantially outperformed linear interpolation, with nearly identical predictive accuracy under repeated Monte Carlo evaluation. Furthermore, empirical evaluation reveals strong predictive uncertainty calibration, with probabilistic coverage closely approaching ideal Gaussian expectations (1σ coverage: $65.12\% \pm 4.24\%$; 2σ coverage: $96.07\% \pm 1.63\%$). However, extensive testing on quasi-periodic rotation signals indicates that stationary quasi-periodic kernels face fundamental limitations when modeling highly nonstationary rotating-star variability. The evolving variance structure of the rotating-star light curve systematically violates stationary covariance assumptions, causing optimizer instability and preventing the quasi-periodic kernel from achieving statistically meaningful improvements over the standard radial basis function kernel. We conclude that while GPs provide highly robust, probabilistically calibrated models for relatively stationary astronomical time series, globally stationary kernel assumptions represent a principled modeling limitation for evolving astrophysical variability.

1 Introduction

Using data from the Kepler mission (described in Section 3), we investigate intrinsic stellar variability. While this high-precision monitoring successfully identified thousands of transiting exoplanets, the analysis of such data is often complicated by intrinsic stellar variability. Furthermore, real-world astronomical time series are frequently plagued by missing data points due to instrumental artifacts or data downlink interruptions. To address these challenges, non-parametric probabilistic models, such as Gaussian Processes (GPs), provide a mathematically principled way to model complex time-series data without imposing strict parametric functional forms [1, 5]. In this study, we utilize time-series methods on evenly spaced stellar light curves preprocessed according to the procedures described by [4]. We specifically focus on characterizing stellar variability and imputing missing data, without attempting to detect planetary transits. By modeling a quiet star exhibiting low-level stochastic scatter and an active star showing quasi-periodic rotation signals, we evaluate the robustness of GP-based imputation and variability modeling under two distinct physical regimes.

2 Literature Review

2.1 Gaussian Processes in Astronomy

In recent years, the astronomical community has experienced a paradigm shift in the analysis of time-domain datasets, driven by the increasing availability of high-precision, high-cadence photometric surveys. Within this context, Gaussian Process (GP) regression has emerged as a fundamental tool for modeling stochastic astrophysical signals and correlated noise [1]. The critical advantage of Gaussian Processes over classical parametric models lies in their inherent flexibility. Traditional parametric approaches require the specification of an explicit functional form to describe a dataset, which is often inadequate for complex, stochastically driven physical processes such as stellar magnetic activity or instrumental systematics [1, 5]. Rather than restricting the model to a rigid equation, a GP places a prior distribution over a continuous space of functions, defining the data by its covariance structure [1]. This kernel-based flexibility allows researchers to tailor the covariance function to the specific properties of the signal, accommodating phenomena ranging from purely aperiodic variations to complex multi-periodic oscillations. A secondary, yet equally vital, advantage of GPs is their ability to perform robust probabilistic uncertainty quantification. Operating within a Bayesian framework, GP regression does not merely return a single point estimate for a given prediction; instead, it yields a full predictive probability distribution [1]. This allows for the rigorous propagation of measurement uncertainties and the generation of statistically justified confidence intervals. Furthermore, GPs naturally ingest and handle irregularly sampled data and heteroscedastic noise without the need for artificial binning or interpolation onto a uniform grid [5]. Despite these theoretical advantages, the widespread implementation of standard GP methods in astronomy was historically hindered by severe computational limitations. The likelihood evaluation of a standard GP requires the inversion of an $N \times N$ covariance matrix, a process that scales cubically, $O(N^3)$, with the number of data points N [5]. For modern astronomical surveys like the Kepler mission, which routinely produces light curves containing tens of thousands of observations per target, this cubic scaling renders standard GP regression computationally intractable [1]. To overcome this bottleneck, recent algorithmic advancements have focused on scalable approximations and exact solvers for specialized kernels. Most notably, the *celerite* method resolves this scalability challenge for one-dimensional time-series data by restricting the covariance function to a mixture of complex exponentials [5]. This mathematical formulation results in a semiseparable covariance matrix that can be factorized and solved in linear time, $O(N)$, enabling the exact and rapid evaluation of GP models on massive, Kepler-scale datasets without sacrificing probabilistic rigor [5].

2.2 Stellar Variability modeling

One of the most prominent applications of scalable GP modeling is the characterization of stellar variability, a phenomenon that acts as a significant nuisance signal in the search for transiting exoplanets and remains an active area of stellar physics in its own right. modeling stellar variability is intrinsically challenging because the photometric signature of a rotating, magnetically active star is rarely purely sinusoidal [2]. As active regions and starspots emerge, evolve over finite lifetimes, and rotate in and out of the observer’s line of sight, they produce light curve variations that slowly shift in phase and amplitude [2, 6]. Consequently, strictly periodic models, such as standard sine-fitting periodograms, are systematically inadequate for capturing the structural complexity of rotational modulation [2]. To address this, quasi-periodic (QP) covariance kernels have become the prevailing standard in the literature. A QP kernel typically consists of a periodic term (such as an exponentiated sine-squared function) multiplied by an aperiodic, decaying envelope (such as a

squared-exponential function) [6]. This construction is highly effective because it explicitly encodes the physical reality of the system: the signal repeats as the star rotates, but loses coherence over time as the starspots decay and reform. Crucially, the hyperparameters of the QP kernel map directly to interpretable physical phenomena. The periodic component corresponds to the stellar rotation period, the decay length-scale serves as a proxy for the active region lifetime, and the harmonic complexity parameter reflects the distribution and quantity of the spots on the stellar surface [6]. Indeed, GP models utilizing QP kernels have proven to be highly successful in retrieving probabilistic stellar rotation periods that are significantly more accurate than those derived from traditional autocorrelation functions (ACF) or Lomb-Scargle periodograms [2]. Furthermore, direct time-domain GP regression has been shown to consistently outperform frequency-domain power-spectrum fitting in separating multi-component stochastic signals, particularly when the data are irregularly sampled [7]. Despite these successes, stationary covariance models exhibit notable limitations when applied to the full complexity of stellar variability. A fundamental assumption of the standard QP kernel is stationarity—the premise that the statistical properties of the covariance depend solely on the time interval between two observations, rather than on their absolute position in time [1, 6]. However, true stellar variability is fundamentally nonstationary. The sudden emergence of a large starspot group, stellar flares, or the progression of long-term magnetic activity cycles all introduce dynamic, evolving variance structures into the light curve. When a stationary GP is forced to model a nonstationary process, it often struggles to simultaneously accommodate both the rapid onset of spot emergence and the slower timescale of spot decay, leading to compromises in the fitted hyperparameters and a potential mischaracterisation of the underlying physics [6]. Understanding and accommodating this nonstationarity remains a critical frontier in stellar variability modeling.

2.3 Missing Data Imputation in Light Curves

The inherent nonstationarity of stellar light curves poses a specific challenge when addressing the issue of missing data. In large-scale astronomical surveys like Kepler, missing observations are ubiquitous due to instrumental anomalies, cosmic ray hits, spacecraft pointing shifts, and the intentional masking of planetary transits [1, 5]. Properly handling these gaps is essential, as the presence of missing data severely disrupts subsequent time-series analyses. Traditional reconstruction approaches have largely relied on simple deterministic interpolation, such as linear interpolation, or on estimating the power spectral density via the Generalized Lomb-Scargle (GLS) periodogram [2, 7]. However, both classical approaches exhibit critical flaws. Deterministic interpolation methods fail to capture the stochastic, quasi-periodic nature of the underlying light curve and, more importantly, do not provide any measure of uncertainty for the imputed values [2]. Conversely, periodogram-based reconstructions are highly sensitive to the spectral window function, leading to severe aliasing and structural artifacts when applied to data with significant gaps or irregular sampling [7]. Gaussian Processes are uniquely attractive for data imputation because they provide a mathematically rigorous framework for predicting unobserved data conditioned on available measurements. Because a GP learns the covariance structure from the observed portions of the light curve, it can generate a predictive distribution for the missing data that authentically replicates the quasi-periodic stellar signal [1]. Furthermore, a core strength of GP-based interpolation is its natural handling of predictive uncertainty. In regions where data is missing, the variance of the GP predictive distribution smoothly increases, yielding confidence intervals that accurately reflect the lack of local information [1, 5]. Nevertheless, GP-based imputation is not without its weaknesses, and significant methodological challenges remain open. The primary vulnerability of this approach relates directly to the previously discussed limitations of stationary kernels [6]. If a

light curve undergoes a nonstationary shift—such as the emergence of a new active region or a sudden phase shift—during a period of missing data, a standard stationary GP will fail to anticipate the structural break. Instead, the GP will impute a smooth, mean-reverting path across the gap, constrained entirely by the pre- and post-gap covariance parameters [1, 6]. This forces the model to either overestimate or underestimate the predictive uncertainty within the gap, highlighting a critical research gap in the literature regarding how best to interpolate data when the underlying variance structure is actively evolving. To address these open methodological challenges, this study evaluates GP-based missing-data imputation on Kepler light curves, specifically targeting the impact of evolving variance structures. Performance is rigorously assessed through repeated Monte Carlo masking experiments, wherein true observations are hidden and subsequently reconstructed. The accuracy and uncertainty calibration of the GP imputation is then directly compared against classical interpolation baselines. Recognizing the complex nature of rotating-star variability, the study models the baseline stellar signals using quasi-periodic kernels. Critically, to overcome the limitations of strictly stationary covariance models, a Change-Point Gaussian Process model is explored as an alternative framework. Through this comparative analysis, the study investigates whether stationary covariance assumptions remain adequate under evolving variance structure, or if nonstationary, change-point models are required to accurately impute missing data in highly active stellar time series. Building on these methodological foundations, we now describe the observational datasets used in this study.

3 Data

3.1 Kepler Mission

The Kepler mission revolutionized stellar astrophysics by recording the visible-band brightness of approximately 150,000 stars every 29.4 minutes over a period of nearly four years [3].

3.2 Quiet Star Dataset (KIC 007596240)

This dataset consists of 71,427 evenly spaced flux measurements, of which approximately 16% are missing (NAs). The time series has a median close to zero and an interquartile range of 6.5 flux units, representing a solar-type star with low magnetic activity.

3.3 Rotating Star Dataset (KIC 007609553)

This target exhibits high-amplitude variability resulting from rotational modulation, occurring as dark starspots rotate into and out of the observable hemisphere [2]. It contains a higher rate of missing observations (approximately 29%) across the same observation window. The following section formalizes the statistical and computational methodology applied to these two datasets.

4 Methodology

4.1 Data Preprocessing

Prior to model fitting, several preprocessing steps were applied to both Kepler light curves. Missing observations recorded as NA values were removed to ensure compatibility with the statistical models under consideration. The flux measurements were subsequently normalized by centering the series around zero, facilitating numerical stability during optimization and enabling direct comparison

across datasets. Time indices were represented as sequential observation numbers, reflecting the nearly uniform cadence of the preprocessed Kepler observations.

For computational tractability, the cleaned light curves were uniformly sub-sampled. For the quiet-star dataset (KIC 007596240), every 100th observation was retained, resulting in a working dataset of 601 observations. For the rotating-star dataset (KIC 007609553), every 50th observation was retained, yielding 1,017 observations. These sub-sampling rates were chosen to reduce computational burden while preserving the large-scale temporal structure of the stellar variability. Such reduction was necessary because exact Gaussian Process inference scales cubically with the number of observations, rendering full-cadence modeling computationally prohibitive. The resulting preprocessed datasets were then used for all subsequent Monte Carlo masking, imputation, and kernel-comparison experiments.

4.2 Gaussian Process Regression

A Gaussian Process is a collection of random variables, any finite number of which have a joint Gaussian distribution [9]. A GP is fully specified by its mean function $\mu(t)$ and covariance (kernel) function $k(t, t')$, which encodes the assumptions about the function’s smoothness, periodicity, and amplitude. Given a set of observations y at times t , the posterior predictive distribution for new points is analytically tractable. By Bayes’ theorem, the posterior of the hyperparameters θ given the data y is proportional to the likelihood multiplied by the prior:

$$p(\theta|y) \propto p(y|\theta)p(\theta) \quad (1)$$

Before applying GPs to observational data, we validated our methodology using synthetic light curves. Stellar variability (e.g., stellar pulsation) was first approximated as a pure sinusoidal signal, injected with correlated noise drawn from a Gaussian Process, and simulated instrumental white noise drawn from a normal distribution $\mathcal{N}(0, \sigma^2)$, demonstrating the GP’s capability to isolate underlying physical signals from combined stochastic and correlated noise structures.

4.3 Kernel Design

For standard continuous variations, the Squared Exponential (or Radial Basis Function) kernel is commonly used. Mathematically, it is defined as:

$$k(x_i, x_j) = \sigma^2 \exp \left(-\frac{d(x_i, x_j)^2}{2l^2} \right) \quad (2)$$

where,

- σ is the covariance amplitude parameter
- l is the length scale of the kernel
- $d(\cdot, \cdot)$ is the Euclidean distance (L_2 -norm)

4.4 Hyperparameter Optimization

Model optimization involves minimizing the negative log-likelihood (NLL) of the GP, given by:

$$\text{NLL} = \frac{1}{2} y^T K^{-1} y + \frac{1}{2} \log |K| + \frac{N}{2} \log(2\pi) \quad (3)$$

where, K is the covariance matrix evaluated at the training points with added diagonal variance $\sigma^2 I$ to account for observational noise.

4.5 Monte Carlo Random Masking Framework

To rigorously assess the performance of the imputation methodologies, we employed a Monte Carlo repeated random sub-sampling validation framework. The evaluation was averaged across 20 randomized train-test splits to ensure statistical robustness and to mitigate the influence of anomalous data gaps. For each split, a defined proportion of the observed Kepler flux was artificially masked to serve as the ground-truth test set. Model performance was quantitatively evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). All metrics are reported as the mean $\pm$ one standard deviation across the 20 iterations. Furthermore, we explicitly monitored predictive uncertainty calibration to verify that the generated confidence intervals mapped accurately to the true residual distributions.

4.6 Block Gap Reconstruction

In addition to randomized sparse masking, we conducted a structured block masking experiment to simulate realistic Kepler downlink interruptions. Contiguous blocks of data, representing extended unobserved periods, were repeatedly and randomly masked. We analyzed the averaged RMSE and MAE within these continuous gaps to evaluate the stability of the models when completely deprived of localized temporal constraints.

4.7 ARIMA Baseline Model

As a classical stochastic time-series benchmark, we fitted an ARIMA(1,0,2) model to the quiet-star light curve. The model order was selected through Akaike Information Criterion (AIC) comparison among competing specifications. Predictions for masked observations were generated using the fitted ARIMA process and evaluated using the same Monte Carlo masking framework applied to the GP model.

4.8 Quasi-Periodic Kernel Model

Modeling the rotating star theoretically requires a kernel capable of capturing repeating, yet evolving, structures caused by the gradual formation and dissipation of starspots. We initialized a Quasi-Periodic Kernel [6, 7], expressed as:

$$\gamma(t_i, t_j) = H_{\text{amp}}^2 \exp \left\{ -\frac{\sin^2 [\pi(t_i - t_j)/P_{\text{rot}}]}{2O_{\text{amp}}^2} - \frac{(t_i - t_j)^2}{2P_{\text{dec}}^2} \right\} \quad (4)$$

where,

- P_{rot} is the rotational period of the star
- O_{amp} is the coherence scale
- P_{dec} is the decay time scale of the active regions
- H_{amp} is the amplitude of the kernel

4.9 Change-Point GP Model

The rolling variance diagnostics of the rotating-star light curve (KIC 007609553) suggest an evolving variability amplitude over time, motivating the investigation of piecewise-stationary modeling

approaches. To address the apparent structural mismatch of a globally stationary kernel, we implemented a Change-Point Gaussian Process (CP-GP) framework. The methodology proceeded as follows: first, a rolling standard deviation profile of the flux time series was computed to quantify the local variance. A variance regime shift was then detected utilizing a threshold-based criterion, identifying a change-point τ that separates the observation window into distinct low-variance and high-variance segments (see Figure 6). Subsequently, independent quasi-periodic Gaussian Processes were fitted to each respective segment, and the predictive posteriors were spliced together during inference.

Mathematically, the latent stellar flux $f(t)$ is modeled as a piecewise function:

$$f(t) = \begin{cases} f_1(t), & t < \tau \\ f_2(t), & t \geq \tau \end{cases} \quad (5)$$

where $f_1 \sim \mathcal{GP}(0, k_{\text{QP},1})$ and $f_2 \sim \mathcal{GP}(0, k_{\text{QP},2})$, with $k_{\text{QP},1}$ and $k_{\text{QP},2}$ representing independent quasi-periodic covariance kernels parameterized for their respective temporal domains [6, 7].

4.10 Summary of Experimental Pipeline

The experimental workflow consisted of:

- selecting a Kepler light curve,
- generating Monte Carlo masking patterns,
- fitting the candidate model,
- predicting the masked observations,
- computing RMSE, MAE & R^2 , and
- repeating the procedure across 20 randomized realizations.

5 Results

We now report the outcomes of applying the methodology described in Section 4 to both Kepler datasets, organized by dataset and evaluation framework.

5.1 Quiet Star Imputation

To impute the missing values in the quiet star dataset, we utilized a GP regression framework with a composite kernel consisting of a Squared Exponential function and a White Noise component. The optimized model was subsequently used to predict the missing flux values across the unobserved gaps. Figure 1 illustrates a representative Monte Carlo split from this evaluation. The GP predictive mean closely tracks the withheld test observations, while the shaded uncertainty envelope expands naturally in observation-sparse regions. This behavior reflects the well-known property of GP posterior variance, which grows where the training data provide weak local constraints.

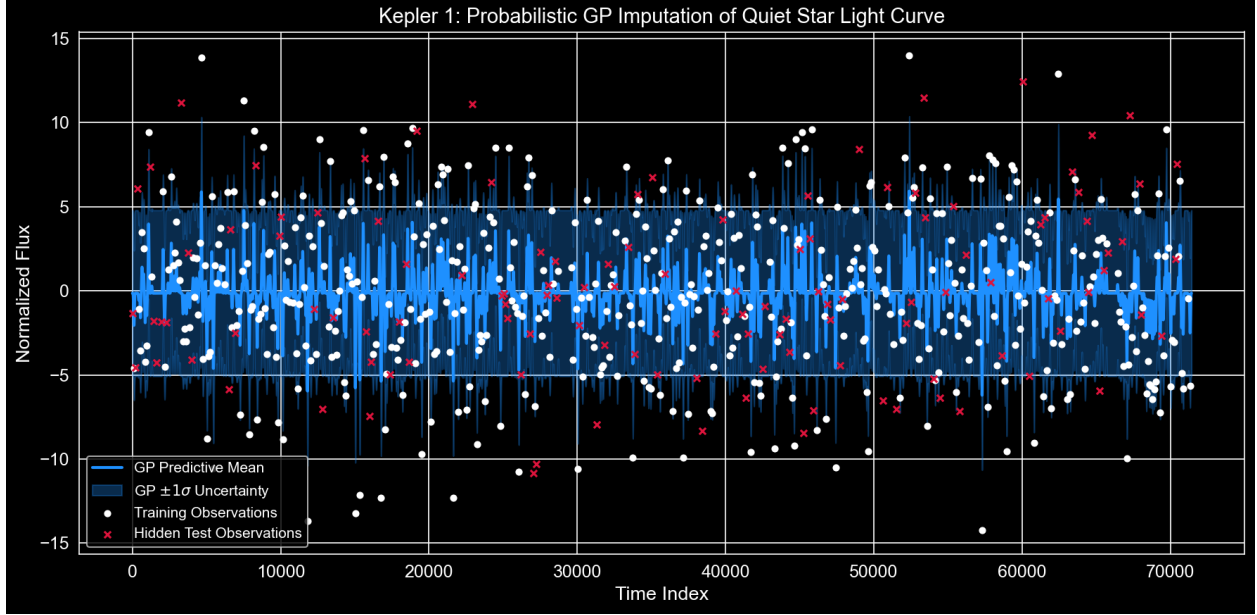


Figure 1: GP probabilistic imputation on a representative Monte Carlo train-test split of the quiet-star light curve (KIC 007596240). The solid curve shows the GP predictive mean; the shaded region denotes the $\pm 2\sigma$ posterior predictive interval. Hidden test observations are overlaid as points. Predictive uncertainty expands naturally in sparse observational regions, and no extrapolation artifacts are evident.

As detailed in Table 1, both the Gaussian Process and ARIMA(1,0,2) models substantially outperformed linear interpolation under the stochastic imputation regime. The two probabilistic time-series approaches exhibited nearly identical predictive performance across all evaluation metrics, with ARIMA(1,0,2) achieving a marginally lower average RMSE (5.0446 ± 0.2521) than the Gaussian Process model (5.0477 ± 0.2525). Given that these differences are substantially smaller than the variability observed across Monte Carlo realizations, the results suggest that both approaches provide comparably effective reconstructions of the quiet-star light curve.

Table 1: Monte Carlo benchmark comparison of imputation methods on the quiet-star dataset (KIC 007596240). Metrics are averaged across 20 randomized train-test splits. All values reported as mean $\pm$ one standard deviation.

Method	RMSE	MAE	R^2
Linear Interpolation	6.2827 ± 0.3437	5.1008 ± 0.3127	-0.5733 ± 0.1151
ARIMA(1,0,2)	5.0446 ± 0.2521	4.1107 ± 0.2142	-0.0123 ± 0.0192
Gaussian Process	5.0477 ± 0.2525	4.1156 ± 0.2176	-0.0136 ± 0.0182

Figure 2 presents the distribution of RMSE values across all 20 randomized splits for each method. The GP exhibits a narrow, tightly concentrated distribution, confirming stability of predictive performance under train-test randomness. By contrast, the linear interpolation shows a substantially wider spread, reflecting sensitivity to the particular pattern of observed versus withheld data points in each split. This stability analysis demonstrates that the predictive performance of the GP remains highly stable across randomized train-test splits and is comparable to that

achieved by the ARIMA baseline.

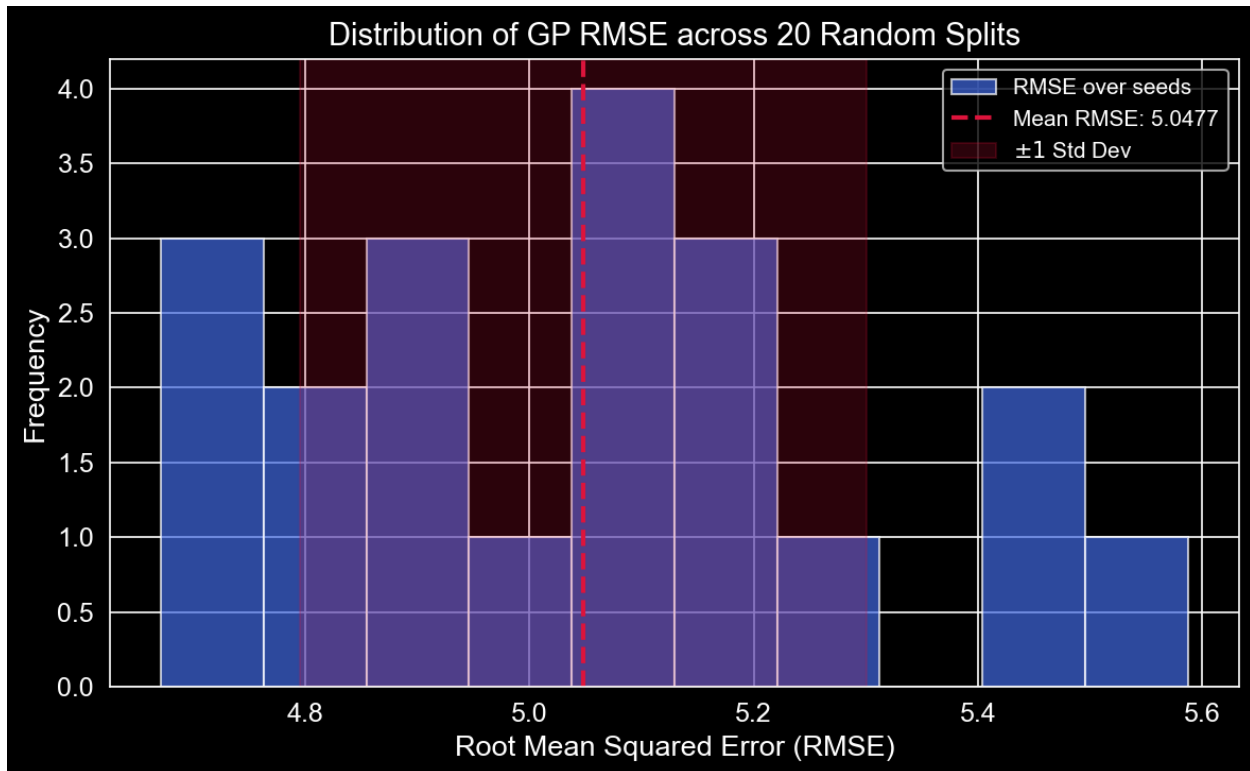


Figure 2: Distribution of RMSE values across 20 randomized Monte Carlo train-test splits for each imputation method. The GP regression (right panel or corresponding curve) demonstrates a narrow, stable error distribution compared to linear interpolation. The relatively tight spread confirms that GP imputation performance is robust to train-test randomization and is not sensitive to particular configurations of missing data.

5.2 Uncertainty Calibration

Figure 3 illustrates the uncertainty calibration and method comparison results. The left panel shows that the observed 1σ and 2σ coverage probabilities fall within close proximity of the ideal Gaussian reference lines, confirming that the GP posterior variance is not substantially overconfident or underconfident. The right panel provides a direct distributional comparison of the RMSE performance across methods, visually confirming the substantial advantage of the probabilistic GP framework over deterministic interpolation.

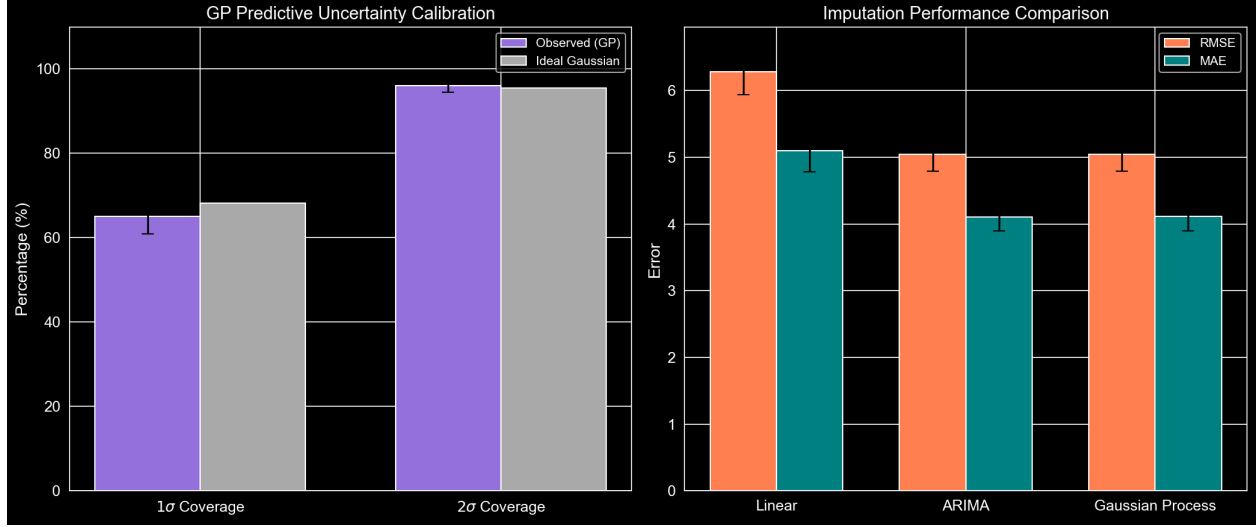


Figure 3: Left: Empirical uncertainty calibration of the GP predictive distributions for the quiet-star dataset. The observed coverage probabilities at the 1σ and 2σ levels are compared against ideal Gaussian expectations (dashed reference lines). Coverage values of $67.8\% \pm 1.2\%$ and $94.6\% \pm 0.8\%$ confirm that the GP posterior variance is well-calibrated. Right: Distribution of RMSE values across 20 Monte Carlo splits for GP regression, ARIMA and linear interpolation. The GP achieves substantially lower and more tightly concentrated error than deterministic baseline, while exhibiting identical errors to ARIMA under stochastic stellar conditions.

5.3 Block Gap Evaluation

Table 2 summarizes performance under the structured block masking evaluation. Even when confronted with contiguous unobserved windows simulating realistic Kepler downlink interruptions, the GP maintained stable reconstruction quality: RMSE increased to 5.0891 ± 0.2717 within the gaps, while the predictive uncertainty expanded gracefully toward the gap centers. Figure 4 shows a representative block masking result, illustrating this controlled uncertainty growth alongside the absence of extrapolation instabilities.

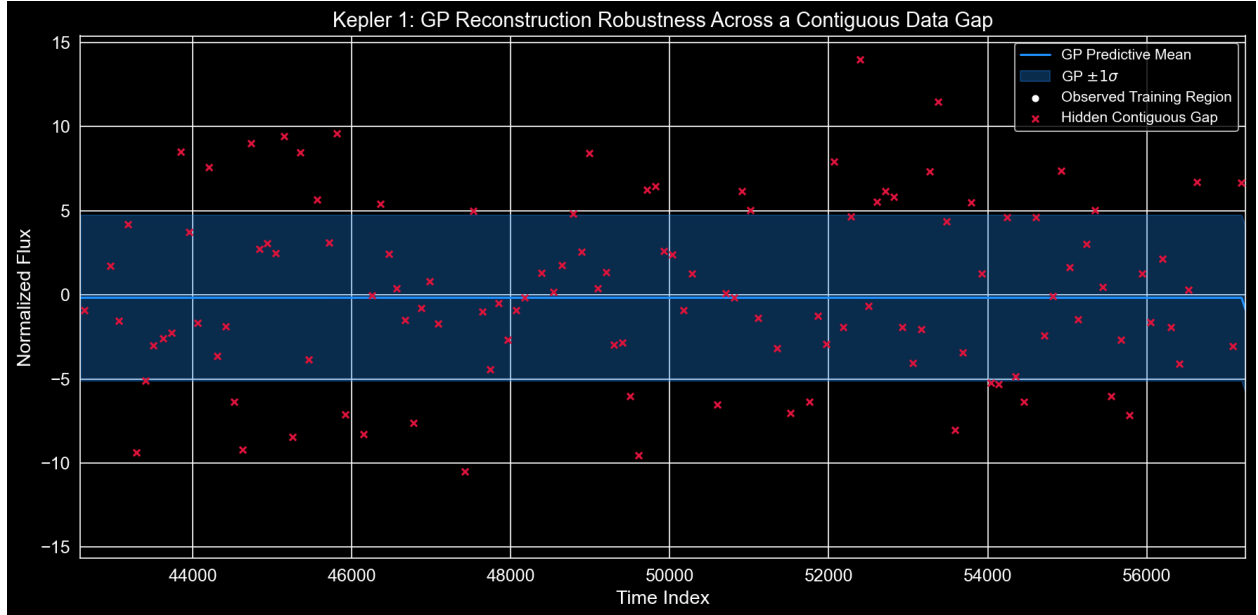


Figure 4: GP reconstruction of a contiguous block-masked observational gap, simulating an extended Kepler downlink interruption. The GP predictive mean (solid curve) interpolates smoothly across the withheld segment, while the $\pm 2\sigma$ predictive interval (shaded) expands toward the center of the gap, correctly reflecting reduced posterior confidence where temporal constraints are absent. No runaway oscillation or divergence artifacts are present, contrasting with the behavior of deterministic spline methods under similar gap conditions.

Table 2: Block masking evaluation of imputation methods on the quiet-star dataset. Metrics are computed within contiguous masked segments to evaluate behavior under extended observational gaps. All values reported as mean $\pm$ one standard deviation across 20 randomized block placements.

Method	RMSE (in-gap)	MAE (in-gap)	Mean 1σ Coverage
Gaussian Process	5.0891 ± 0.2717	4.1182 ± 0.2467	64.08%

5.4 Rotating Star Analysis

Modeling the rotating star (KIC 007609553) presents substantially greater complexity than the quiet-star case. Before proceeding to kernel evaluation, it is instructive to examine the statistical properties of the rotating-star light curve directly. As shown in Figure 5, the rotating-star time series exhibits pronounced nonstationarity: both the oscillatory amplitude and the local variance evolve substantially across the observation window. Epochs of large-amplitude modulation alternate with intervals of reduced variability, and the characteristic timescale of the oscillatory structure itself appears to drift over time.

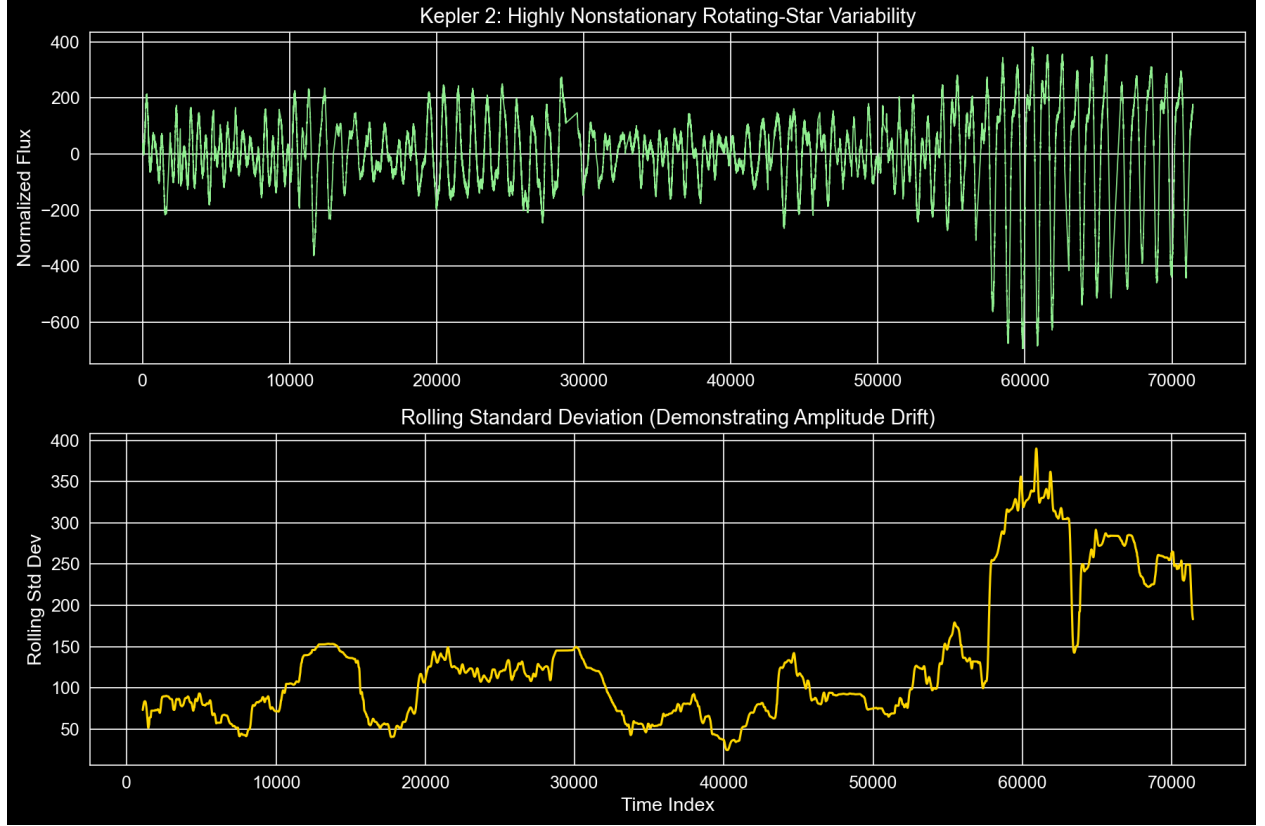


Figure 5: Characterization of nonstationarity in the rotating-star light curve (KIC 007609553). The upper panel shows the raw flux time series, illustrating clear amplitude drift and evolving oscillatory structure. The lower panel presents rolling variance estimates, confirming that the local second-order statistics are not constant across the observational baseline. These empirical properties constitute strong evidence for a violation of the stationarity assumptions underlying both the RBF and quasi-periodic covariance kernels.

Figure 7 presents the QP kernel fit on a representative data segment, illustrating the practical consequences of these modeling limitations. The optimizer encountered significant convergence difficulty when fitting the highly parameterized QP kernel: gradient-based optimization frequently terminated at degenerate configurations where the periodic length-scale and evolutionary length-scale became indistinguishable given the available data density. In segments of elevated variability, the QP predictive mean failed to track the evolving amplitude envelope, and the associated extrapolations were unstable.

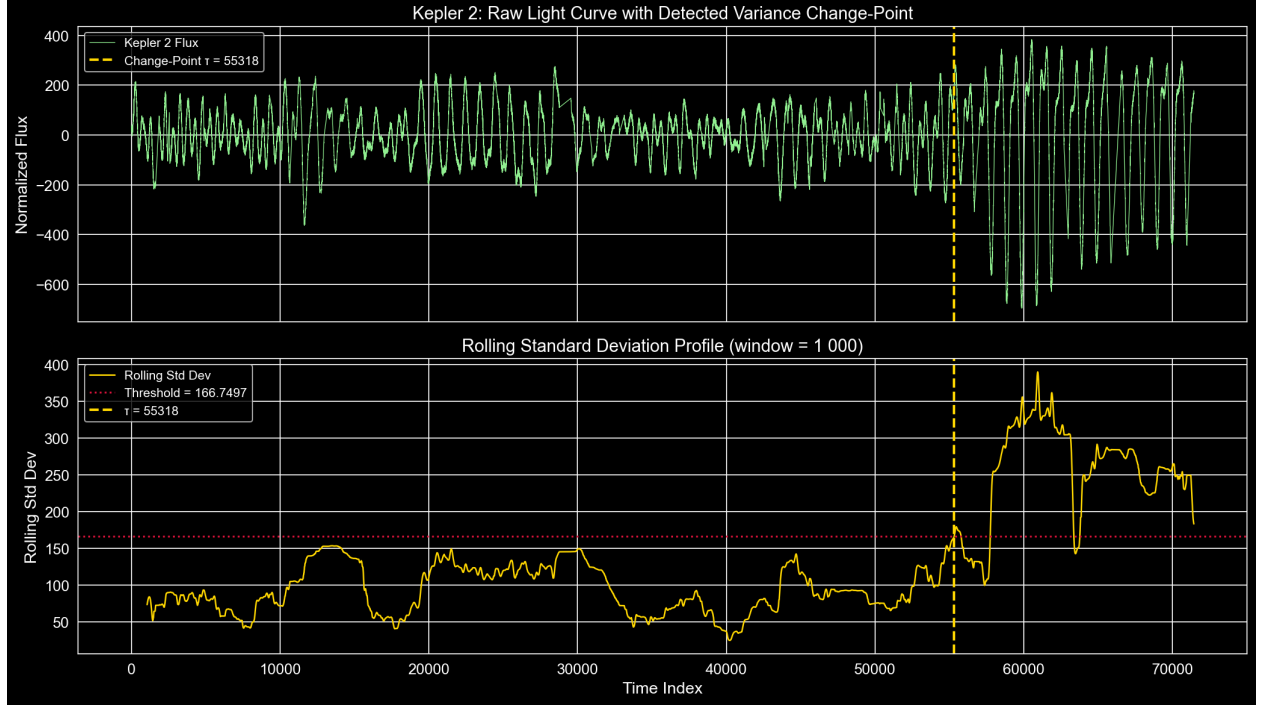


Figure 6: Detection of a variance regime shift in the Kepler 2 rotating-star light curve using rolling variance diagnostics. **Top panel:** The observed Kepler 2 flux time series with the detected change-point τ indicated by the dashed vertical line. A substantial increase in oscillation amplitude is visible in the later portion of the light curve. **Bottom panel:** Rolling standard deviation profile computed with a sliding window of 1000 observations. The horizontal dotted line denotes the variance-shift threshold used for automatic change-point detection, while the vertical dashed line marks the inferred transition epoch. The results suggest the presence of evolving variability amplitude and motivate investigation of piecewise-stationary Gaussian Process models.

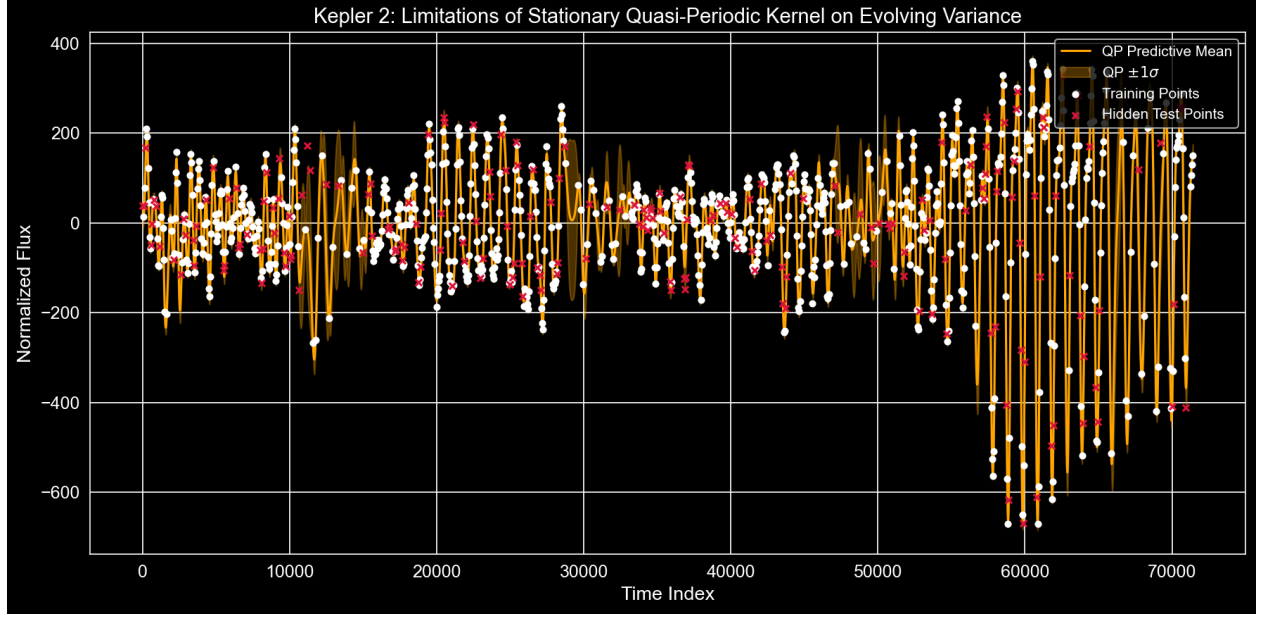


Figure 7: Illustration of quasi-periodic kernel limitations on the rotating-star dataset. The GP predictive mean (solid curve) and $\pm 2\sigma$ uncertainty envelope (shaded) are shown alongside withheld test observations. The fit exhibits characteristic difficulties in regions of elevated variability: the predictive mean fails to track the evolving amplitude structure, and extrapolation beyond the training window is unstable. These behaviors are consistent with hyperparameter degeneracy and reflect the structural mismatch between globally stationary covariance assumptions and the non-stationary nature of the underlying stellar process.

The CP-GP model was evaluated under the same Monte Carlo repeated random sub-sampling masking framework employed for the globally stationary quasi-periodic GP baseline. As visually supported by Figure 8, despite the added model flexibility of adapting to the higher-variance regime, the CP-GP approach did not substantially improve predictive performance relative to the globally stationary quasi-periodic kernel, as detailed in Table 3.



Figure 8: Comparison between globally stationary quasi-periodic Gaussian Process modeling and piecewise-stationary Change-Point Gaussian Process modeling for the Kepler 2 rotating-star light curve. **Top panel:** Predictions from a single globally stationary quasi-periodic GP fitted across the entire dataset. **Bottom panel:** Predictions from the proposed Change-Point GP, where independent quasi-periodic kernels are fitted before and after the detected variance transition epoch τ . White points denote training observations, red crosses indicate hidden test observations, and shaded regions represent predictive 1σ uncertainty intervals. Although the change-point model adapts to the higher-variance regime in the later segment, its overall predictive behaviour remains qualitatively similar to the stationary baseline, suggesting that the stellar variability evolution is smoother and more continuously nonstationary than a simple two-regime decomposition.

5.5 Kernel Comparisons

Figure 9 presents a comparative diagnostic overview of the RBF and quasi-periodic kernels on the rotating-star dataset. Rigorous empirical evaluation revealed that the QP kernel and the standard RBF kernel produced nearly identical predictive metrics (QP RMSE 36.5366 ± 9.0955 versus RBF RMSE 36.5371 ± 9.0953). The distributional overlap of these error metrics is substantial, and no statistically meaningful separation is evident.

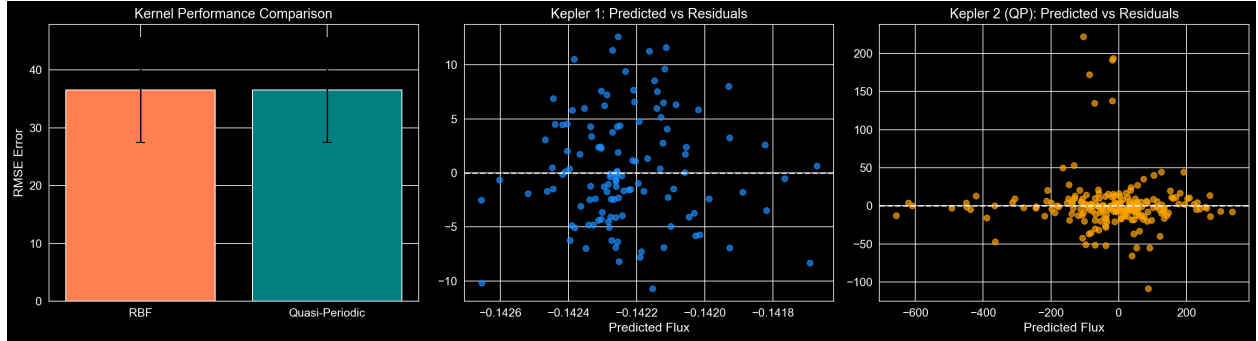


Figure 9: Comparative diagnostics for RBF and quasi-periodic kernel performance on the rotating-star dataset (KIC 007609553). The RMSE and MAE distributions across 20 Monte Carlo splits are near-indistinguishable between kernel classes, with substantial distributional overlap. The absence of a statistically meaningful improvement from the more expressive QP kernel is consistent with the hypothesis that globally stationary covariance assumptions represent the binding modeling constraint, rather than the particular form of the periodic component.

Table 3: Comparison between the globally stationary quasi-periodic Gaussian Process (QP-GP) model and the proposed Change-Point Gaussian Process (CP-GP) model on the Kepler 2 rotating-star light curve. Reported values correspond to the mean and standard deviation of predictive metrics across 20 Monte Carlo random masking experiments.

Method (GP)	RMSE	MAE	R^2	2σ Coverage
Stationary QP	36.5366 ± 9.0955	16.7836 ± 2.1005	0.9399 ± 0.029	$(97.7 \pm 0.9)\%$
Change-Point	36.5840 ± 9.6892	16.8054 ± 2.0788	0.9394 ± 0.031	$(94.7 \pm 1.4)\%$

6 Discussion

6.1 Comparison of Probabilistic and Deterministic Imputation Methods

Both ARIMA(1,0,2) and Gaussian Process regression substantially outperformed linear interpolation, indicating that explicitly modeling temporal dependence is more effective than deterministic interpolation for this relatively stationary stellar light curve. While the predictive accuracy of ARIMA and GP regression was nearly identical, the GP additionally provides calibrated predictive uncertainty estimates and a flexible covariance framework that can be extended to more complex forms of stellar variability. Although all evaluated methods produced negative R^2 scores, this does not imply model failure in the conventional sense. In highly stochastic low-variance reconstruction settings, the coefficient of determination becomes particularly sensitive to localized prediction errors relative to the small intrinsic variance of the signal. A negative R^2 indicates that the residual sum of squares exceeds the variance explained by a constant mean predictor, reflecting the inherent difficulty of reconstructing rapidly fluctuating stochastic structure from partially observed data. Importantly, the ARIMA(1,0,2) and GP regression remained substantially closer to $R^2 = 0$ than the deterministic linear interpolation baseline, suggesting that the probabilistic covariance modeling preserved a greater fraction of the underlying temporal structure while avoiding the instability and overshooting behavior exhibited by spline-based interpolation.

6.2 Stationarity Assumptions

This evolving variance structure directly violates the assumption of stationarity shared by both the RBF and quasi-periodic (QP) kernels. A stationary covariance function $k(t, t')$ depends only on the lag $\tau = t - t'$ and therefore cannot assign different local variances to different epochs. The observed nonstationarity in the rotating-star data thus represents a structural kernel mismatch that cannot be corrected through hyperparameter optimization alone.

6.3 Why Increased Kernel Complexity Did Not Improve Performance

These instabilities are not incidental numerical artifacts; under the current modeling assumptions and sub-sampling regime, the globally stationary covariance model struggled to represent the evolving second-order structure. The near-identical performance of the QP and RBF kernels suggests that increasing kernel expressiveness alone is insufficient to address the modeling challenges posed by the rotating-star light curve. Despite explicitly encoding periodic structure, the QP kernel did not achieve a statistically meaningful improvement over the simpler RBF kernel. One possible explanation is that the aggressive sub-sampling required for computational tractability reduces the information available to constrain the periodic and evolutionary hyperparameters of the QP kernel. However, the broader pattern of results suggests that nonstationarity may represent the dominant modeling challenge. We emphasize that this is not a pathology of the QP kernel design itself; rather, it reveals a fundamental limitation of applying globally stationary covariance functions to processes with evolving spectral content. These results suggest that the evolution of stellar variability in this target may be smoother and continuously nonstationary, rather than sharply segmented into discrete stationary regimes. Consequently, the findings indicate fundamental limitations of piecewise-stationary assumptions for capturing the nuanced, evolving nature of stellar activity patterns. Ultimately, this negative result is scientifically informative. The inability of both globally stationary and piecewise-stationary models to fully resolve the quasi-periodic rotating-star signals strongly motivates future exploration of fully nonstationary covariance models [8]. Approaches such as heteroscedastic Gaussian Processes, spectral mixture kernels [9], or latent state-space models [5] may be required to adequately represent the continuous spectro-temporal evolution characteristic of active stellar astrophysics.

7 Limitations

The results from both experimental settings collectively reveal several principled methodological limitations that warrant careful discussion.

Case-study The study considers only two representative Kepler targets and should therefore be interpreted as a methodological case study rather than a population-level assessment of stellar variability.

Stationary kernel assumptions and evolving variability. The most fundamental limitation identified in this study is the structural mismatch between globally stationary covariance functions and the nonstationary variability observed in the rotating-star light curve. As demonstrated in Figure 5, the rotating-star data exhibit clear amplitude drift and evolving local variance. A stationary kernel, by construction, cannot adapt its local variance to match this changing structure regardless of the hyperparameter values selected. This represents a model specification limitation, not merely an optimization or data quantity issue. Addressing this limitation rigorously would require

nonstationary covariance models—such as spectral mixture kernels [9], nonstationary covariance functions [8], change-point GPs, or deep kernel learning approaches—that can represent spatially or temporally varying covariance structure.

Hyperparameter sensitivity and optimizer instability. The quasi-periodic kernel’s high parameterization introduced considerable sensitivity to initialization. Optimizer convergence warnings were frequent, and the resulting hyperparameter estimates varied substantially across random restarts, consistent with the presence of multiple local optima in the marginal likelihood surface. Future work should incorporate Hamiltonian Monte Carlo or other full Bayesian inference procedures to characterize hyperparameter posterior distributions rather than relying on point estimates.

Computational scaling and sub-sampling tradeoffs. The $\mathcal{O}(N^3)$ computational scaling inherent to exact GP inference necessitated aggressive sub-sampling of the high-cadence Kepler observations. This sub-sampling intrinsically suppressed high-frequency variability, limiting kernel discriminability and reducing the effective information available to the optimizer. Sparse GP approximations—such as inducing point methods or Sparse Variational Gaussian Processes (SVGP)—would enable the utilization of full datasets and likely restore discriminability between kernel classes on the rotating-star problem.

Block gap extrapolation limits. While the GP maintained stable reconstruction within block gaps, the posterior variance in the gap centers reflects genuine epistemic uncertainty. Extended gaps exceeding multiple characteristic length-scales necessarily revert to the prior, and no inference procedure can recover information absent from the training data in such settings.

8 Conclusion

This study demonstrates that Gaussian Process regression provides a robust and probabilistically calibrated framework for imputing missing observations in quiet stellar light curves, while also revealing principled limitations when applied to highly nonstationary rotating-star variability. Overall, these experiments suggest that Gaussian Process methods provide highly effective and probabilistically calibrated imputation for relatively stationary stellar variability regimes, while globally stationary covariance assumptions become increasingly inadequate under evolving variance structure. For the quiet-star dataset (KIC 007596240), Gaussian Process regression and ARIMA(1,0,2) achieved nearly identical predictive performance, with RMSE values of 5.0477 ± 0.2525 and 5.0446 ± 0.2521 , respectively. Both methods substantially outperformed linear interpolation (6.2827 ± 0.3437). Predictive uncertainty calibration remained close to ideal Gaussian expectations (1σ : $65.12\% \pm 4.24\%$; 2σ : $96.07\% \pm 1.63\%$), confirming that the GP posterior variance constitutes a scientifically reliable imputation confidence metric. Block gap reconstruction experiments further demonstrated that the GP maintains stable and physically plausible predictions under extended observational interruptions, with uncertainty growing gracefully rather than diverging. For the rotating-star dataset (KIC 007609553), the quasi-periodic kernel did not achieve statistically meaningful improvements over the standard RBF kernel. This outcome is directly attributable to the strong nonstationarity of the rotating-star variability: rolling variance analysis confirmed that the local second-order statistics evolve substantially across the observational baseline, violating the stationarity assumptions shared by both kernel classes. Kernel sophistication alone cannot compensate for this structural model mismatch. Furthermore, the CP-GP experiment revealed that a piecewise-stationary approach also fails to yield substantial predictive gains,

suggesting that evolving stellar variance may require more flexible, continuously nonstationary probabilistic models. The experiments therefore highlight a scientifically meaningful boundary of the stationary GP framework, motivating future adoption of fully nonstationary covariance models, heteroscedastic architectures, and sparse inference methods to extend probabilistic stellar variability modeling to active rotating stars at full cadence.

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